

Rebuttal to Reviewer GBAF

CoLM 2026 — Submission 991

Dear Reviewer GBAF,

We sincerely thank you for the careful and supportive review, and we are especially grateful that you highlighted the ranking-vs.-reliability finding in **Section 5.2** as adding depth to how high-performing judges with strong correlations can still produce poor point score estimates. We are also glad the intentional choice of three heterogeneous judges (a specialist judge, a larger reasoning model, and a closed-source generalist) came through as a deliberate design rather than an arbitrary sample. Your two specific suggestions — a more formal task taxonomy and citations for the conformal methods — are both clear improvements that we will incorporate. We address each below and end with a short summary of camera-ready revisions.

R1. A more formal task taxonomy

You are right that the current 4-way grouping (Vision-Heavy / General VQA / Knowledge-Web / Aesthetics-AI) is informal, and the COCO-vs-TextVQA placement you flagged is a clear example of the inconsistency. We will adopt a principled taxonomy drawn from existing work.

Specifically, we will re-categorize the 14 MLLM-as-a-Judge tasks using the **6 vision capability dimensions** from MM-Vet (Yu et al., 2023): Recognition, OCR, Knowledge, Language Generation, Spatial Awareness, and Math reasoning. MM-Vet is a widely-cited multimodal benchmark whose taxonomy was constructed explicitly to disentangle these capabilities. Under MM-Vet, the exact placement you flagged resolves cleanly: **COCO** (natural-image captioning) maps to *Recognition* and *Language Generation*, while **TextVQA** (reading text from images) maps to *OCR*. The two tasks are then separated by the capability they actually exercise rather than by our informal labels, and the apparent arbitrariness in placing one as “General VQA” and the other as “Vision-Heavy” disappears. Other tasks map similarly: MathVista to *Math*, ChartQA and InfographicsVQA to *OCR + Math*, ScienceQA to *Knowledge*.

To avoid double-counting tasks that span capabilities, we will assign each task a *primary* capability (used for the per-task width row in Table 3 so each task appears once with one width) and a secondary-capabilities tag column for transparency. We will replace the current Table 3 grouping with the MM-Vet primary-capability grouping and rewrite §5.4 to discuss the task-dependent uncertainty pattern under this principled taxonomy. We will report the resulting per-capability width pattern transparently rather than predicting its shape, and state in the text how the original task-dependent trend (Table 3, current grouping) appears under the formal grouping. We will cite MM-Vet (Yu et al., 2023) as the source. The per-task widths cited elsewhere in the paper (e.g., Table 3 individual rows; the per-dataset RSG analysis in §5.5) are unchanged — only the grouping of rows is re-labeled.

Q1. Citations for the conformal methods + compressing §5.1

You are correct that §5.1 currently lists method names without inline citations, which assumes more conformal-prediction familiarity than we should. We will add a citation immediately after each method name. The intended bibliography entries are:

- **Naive Split CP** — Vovk et al. (2005)
- **CQR (Conformalized Quantile Regression)** — Romano et al. (2019)
- **Asymmetric CQR** — Sesia & Candès (2020)
- **CHR (Conformal Histogram Regression)** — Sesia & Romano (2021)
- **LVD (Locally-adaptive Variance-based Distribution-free)** — Lin et al. (2021)
- **Boosted CQR / Boosted LCP** — Xie et al. (2024)
- **OrdinalAPS (Ordinal Adaptive Prediction Sets)** — Lu et al. (2022)
- **R2CCP (Regression-as-Classification CP)** — Guha et al. (2024)

Most of these are already in our bibliography for other citations elsewhere in the paper; we simply did not surface them inline in §5.1. We note also that Boosted CQR and Boosted LCP are two variants from the same source paper (Xie et al., 2024), which is why the paper refers to “8 conformal methods” while Table 2 enumerates 9 rows (the two Boosted variants are counted as one family); we will clarify this in the §5.1 prose so the count is unambiguous.

We also take your “this contributes less to the overall narrative and could probably be moved to the conclusion” suggestion seriously. Rather than the conclusion itself, we think the natural home for the full per-method detail is the appendix, which keeps the main text focused on the empirical findings while preserving the methodological content for readers who want it; we are happy to move it to the conclusion instead if you prefer. Concretely, in the camera-ready we will **compress §5.1’s main-text prose** — keeping only a concise summary of which methods we evaluate and why R2CCP and CHR are our headline choices — and **move the full per-method descriptions and comparison details into the appendix**. This will tighten the main-text narrative around the empirical findings (which you noted as the paper’s contribution) without losing the methodological detail for readers who want it.

Summary of camera-ready revisions

In response to your review we will make the following changes:

1. **§5.4 / Table 3** — replace the informal 4-way task grouping with the MM-Vet 6-capability taxonomy (Yu et al., 2023); re-aggregate the per-task uncertainty results under this principled taxonomy and revise §5.4 prose accordingly.
2. **§5.1** — add inline citations after every conformal-method name (Naive Split, CQR, Asymmetric CQR, CHR, LVD, Boosted CQR, Boosted LCP, OrdinalAPS, R2CCP); compress the main-text prose so the section is a tight summary, and move the full per-method comparison details into the appendix.

We are very grateful for the supportive review and for the two specific, actionable suggestions, which directly tighten the parts of the paper that needed it most. We remain available throughout

the discussion period to answer any further questions or share intermediate materials if helpful.